Dell Automation Platform 1.*x*

Deployment Guide

September 2025
Rev. A01

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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Introduction

Topics:

- Overview
- Terminology

Overview

This document describes the software and procedures that are required to deploy Dell Automation Platform components into one of the following on-premises environments:

- A single-node Kubernetes cluster for Helm-based deployments
- A three-node Kubernetes cluster for Helm-based deployments
- A vCenter cluster managed by VMware vSphere Web Client
- An ESXi cluster managed by VMware vSphere Web Client

Deployments for Dell Private Cloud and Dell AI Solutions require ProDeploy or ProDeploy Plus. For more information, see the Dell ProDeploy Enterprise Suite Service Description.

SaaS customers do not install Dell Automation Platform. If you have a SaaS environment, go to the Dell Automation Platform SaaS portal instead.

For support, go to the Dell Automation Platform support site.

NOTE: Many PDF viewers add a line break to the end of each line of text in a PDF. Adobe Acrobat and other common PDF viewers, including Google Chrome and Microsoft Edge, insert these line breaks. As a result, when you copy code that wraps across multiple lines in a PDF, the copied code contains erroneous line breaks. If you paste the code, the line breaks can cause syntax errors.

To address this known limitation and ensure that copied code does not contain unintentional line breaks, do one of the following:

- Paste the copied code into a text editor and remove the line breaks.
- Use the HTML version of this document when you copy code.

Prior to deploying Dell Automation Platform, it is essential that you complete the steps described in the *Dell Automation Platform 1.x Pre-deployment Guide* available on the Dell Automation Platform support site.

Terminology

The following terms are used throughout this document.

Table 1. Terminology

Term	Definition
Block storage provider	Block storage divides stored data into blocks of equal sizes in a manner that is optimized for fast access and retrieval. This example uses Longhorn as a block storage provider, which also comes with backup and snapshotting capabilities.
CNI	A Container Network Interface (CNI) is a network configuration format for administrators. It contains directives for both the container runtime and the plugins to consume. At plugin execution time, the runtime interprets this configuration format and transforms it into a form to be passed to the plugins.
Container registry	A repository to store container images used by a container runtime.

Table 1. Terminology (continued)

Term	Definition
CSI	A Container Storage Interface (CSI) is a standard for exposing arbitrary block and file storage systems to containerized workloads in Kubernetes to make the volume layer extensible. Third-party storage providers can deploy plugins to expose new storage systems without modifying the core Kubernetes code, which gives users more options for storage and makes the system more secure and reliable.
Dell Automation Platform infrastructure	Select Dell hardware that is optimized for the platform to provide a secure and scalable operating environment to run applications as VMs and containers.
Dell Automation Platform orchestrator	A software platform for orchestration of edge devices and applications.
Dell Automation Platform installation bundle	A ZIP file containing all software and image files that are required for installation of the Dell Automation Platform components.
Dell Automation Platform portal	This is the component of platform hosted services such as MFEs, org, user, roles, asset, policy and IAM components. The portal provides a centralized place for customers to manage their org/users.
Dell Automation Platform proxy	An application that partitions the connection between client and server. The proxy accepts certain types of traffic entering or leaving a network, and processes and forwards it.
Docker	A container runtime and a collection of command-line utilities for managing containers and container images. When installed using Kubernetes, the Dell Automation Platform installation process uses Docker to manage several container images.
Helm	A package management system for Kubernetes. Helm allows administrators to build templates for Kubernetes resources and manage the life cycle of an installation. The Dell Automation Platform installation process deploys numerous resources using Helm charts.
High-level container runtime (`containerd`)	A software component that runs on top of a low-level container runtime to provide additional functionality.
Kubernetes	A container orchestration engine. Kubernetes, also known as K8s, is a supported platform for deploying Dell Automation Platform.
Load balancer	A load balancer distributes tasks over a set of resources to improve overall processing efficiency. A load balancer optimizes response time and avoids unevenly overloading some compute nodes while other compute nodes are left idle.
Low-level container runtime (`runc`)	A container engine that implements the Open Container Initiative (OCI) runtime specification. Low-level runtimes are the basic components that allow containers to function, and they perform tasks like creating, starting, stopping, and unpacking containers.
Orchestrator proxy	A server that communicates with SaaS orchestrator and onboarded machines. Using a proxy server, users can deploy blueprints to the onboarded devices from the SaaS orchestrator.
OVA	An Open Virtualization Appliance file is a single file that archives all the files that make up an Open Virtualization Format (OVF) package.
OVF	The Open Virtualization Format (OVF) is an open standard for packaging and distributing virtual appliances.
SaaS	Software as a Service (SaaS) is a capability that is provided for the consumer to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through a thin client interface, such as a web browser or a program interface. The consumer does not manage or control the underlying cloud infrastructure except for limited user-specific application configuration settings.

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Revision history

Table 2. Revision history

Revision	Date	Description of changes
A01	Sep 2025	• Updated Overview. • Updated SaaS licensing. • Updated Access configuration requirements for deployment • Updated Deploy the Dell orchestrator proxy into Kubernetes with Helm. • Removed *Alternative deployment methods using a public repository*. • Updated Access configuration requirements for deployment and Day-2 operations • Updated Initializing the Dell Automation Platform portal and orchestrator for on-premises deployments.
A00	Aug 2025	Initial release. Some content is merged from the *Dell NativeEdge Orchestrator 3.1 Deployment Guide*.

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Enabling the Dell Automation Platform portal and orchestrator for SaaS

If you're a SaaS user, you first initialize the Dell Automation Platform portal and orchestrator and then deploy a Dell orchestrator proxy. If you plan to use the Dell Automation Platform on-premises, there is no need to install the orchestrator proxy, and you can proceed to Deploying the on-premises Dell Automation Platform.

Topics:

- Initializing the Dell Automation Platform portal and orchestrator for SaaS deployments
- Deploying the SaaS Dell orchestrator proxy

Initializing the Dell Automation Platform portal and orchestrator for SaaS deployments

This section describes the initial setup steps for the Dell Automation Platform SaaS portal and orchestrator, along with suggestions to ensure that your Dell Automation Platform account is entitled and ready for immediate use.

Initialize the Dell Automation Platform for the first time

About this task

When you first log into the Dell Automation Platform portal, you must complete a series of one-time steps to initialize the service:

Steps

1. Read the Dell Automation Platform Terms and Conditions, select the checkbox that you have read and agree to all of the terms and conditions, and click **Accept**.

   After you accept the agreement, the Dell Automation Platform automatically downloads ownership vouchers from Dell MyAccount for hardware devices that support it. This enables automated and secure onboarding of these applicable devices into Dell Automation Platform. As vouchers download to your portal, the associated devices are displayed in the **Assets** page of the portal.

   NOTE: Device vouchers might become available for download later, as new Dell hardware is ordered and shipped to your location. Agreeing to this one-time prompt allows all current and future vouchers to download automatically, enabling automatic onboarding of the associated hardware devices.

2. To activate your Dell Automation Platform orchestrator instance, navigate to your orchestrator using the main portal display or when prompted in the **Portal Assets** screen.

   When you navigate to your SaaS-based orchestrator for the first time, it is licensed from an available entitlement that is associated with your Dell Automation Platform account. If you do not have an entitlement to use the Dell Automation Platform SaaS orchestrator, you are prompted to provide a license before proceeding. Contact Dell Sales to purchase an entitlement, or contact Dell Support to identify an existing entitlement.

   If you have purchased NativeEdge devices, you have already agreed to allow voucher downloads. If your orchestrator is properly licensed, you see an alert informing you that an orchestrator administrator must enter a NativeEdge Escrow Key Passphrase before NativeEdge devices can onboard to the orchestrator. Click on the link in the alert and then follow the on-screen instructions.

3. To create an **Encryption Key Escrow Passphrase**, hover over the information icon for details about complexity requirements.

WARNING: It is critical that you take note of the Encryption Key Escrow Passphrase now. It cannot be retrieved later, and it will be required for future restore and replacement tasks.

NOTE: Note: If you do not yet own NativeEdge devices or have not downloaded their associated vouchers to your Dell Automation Platform Portal, you will not see the alert regarding the passphrase. However, you can still provide a passphrase if you anticipate onboarding Dell Automation Platform devices in the future. To do so, go to the orchestrator and navigate to the **Security** page under the **Administrator** side navigation pane. Next, navigate to the **Escrow Passphrase** tab, and follow the on-screen instructions to create an acceptable passphrase.

SaaS licensing

The Dell team provisions the SaaS portal and orchestrator. If the user exists in Dell Identity with the proper entitlement, the SaaS licensing is applied automatically.

Each user who requires access to license entitlements must ensure that their Dell Identity profile is correctly linked to their organization profile in the Customer Profile Data (CPD) system.

This is important because if the user's Dell Identity profile is not linked to the correct organization, they will not see any license entitlements even if they are authorized.

If a user is unable to view license entitlements, raise a service request from within Dell Automation Platform.

As a best practice, include Dell Identity profile validation as part of your user-provisioning checklist.

Access configuration requirements for deployment

To ensure entitlement visibility, licensing accuracy, and compliance with Dell licensing policies, the following access configurations are mandatory during both initial deployment and ongoing operations.

Access key and PIN configuration

The access key and PIN are used to configure connectivity to Dell for both SaaS and on-premises Dell Automation Platform deployments. In SaaS, they are used for Dell Support only, while on-premises deployments use them for support and licensing operations.

CAUTION: The access key and PIN used to configure Dell Connectivity Services must belong to the Administrator account.

Credentials from deployment engineers or temporary users must not be used. Other methods to configure the access key and PIN are strictly prohibited.

Understanding how access keys and PINs function is important because the Dell Automation Platform orchestrator retrieves license entitlements based on the identity that is linked to the access key and PIN. If incorrect credentials are used, entitlements may be mapped to the wrong organization. This can lead to several issues, including loss of entitlement visibility for the customer, data integrity problems, and failures in compliance reporting.

CAUTION: Compliance note—Any attempt to bypass this requirement violates licensing policy and may result in audit failures or support limitations.

For details on how to generate an access key and PIN, see Generate an access key on the Dell support site.

Hierarchy validation for SaaS deployments

For Dell Connectivity Services to function correctly, the administrator whose access key is used must be positioned within the hierarchy so that they have access to the required entitlements. This requirement is critical because entitlement visibility within the Dell Automation Platform is dictated by the administrator's position within the hierarchy. They must have access to entitlements that were purchased for a primary location in the hierarchy, as well as any entitlements purchased for child locations. The user does not have access to any entitlements purchased for a location that is positioned above the administrator user's location or in a separate branch in the hierarchy.

Misalignment can result in incomplete license provisioning, missed renewals, and operational blind spots. To prevent these issues, it is essential to confirm the CPD hierarchy before deployment—either by using internal tools or by contacting Dell Support. Additionally, ensure that the access key owner has full visibility into all entitlements across the customer's organization.

CAUTION: Compliance risk—Misalignment in the hierarchy can result in partial license visibility, which may affect compliance reporting and support eligibility.

Deploying the SaaS Dell orchestrator proxy

You can deploy the SaaS Dell Automation Platform orchestrator proxy either as a Helm installation into an existing Kubernetes environment or as an OVA into an existing VMware vCenter or ESXi setup.

The Dell Automation Platform orchestrator proxy serves as a protective layer for on-premises customer devices, preventing direct exposure to the Internet while enabling secure connectivity to the SaaS Dell Automation Platform orchestrator. It safeguards identities by routing traffic between on-premises devices and the Dell-hosted SaaS orchestrator.

Note that the Dell Automation Platform orchestrator proxy is not required if the only on-premises devices being used with the Dell Automation Platform are NativeEdge endpoints.

NOTE: Dell Technologies expects customers to install their Kubernetes environments and to ensure compliance with the requirements in this document before the Dell orchestrator proxy deployment. Customers who want assistance to perform those tasks should contact their Dell sales representative.

NOTE: These steps are for SaaS deployments only and do not apply to on-premises deployments.

The following minimum hardware requirements for the Dell orchestrator proxy must be met:

CPU	8 CPU cores
Memory	16 GB RAM
Hard disk	1 TB disk space (thick provisioned)
Network	10 Gbps

In SaaS environments, Dell Automation Platform administrators may require the ability to:

- Use the Catalog on a Dell Automation Platform orchestrator hosted either by Dell or in a public cloud to deploy blueprints.
- Deploy and manage assets that are in their organization's private data center securely without exposing their devices directly to the Internet.

The Dell orchestrator proxy enables administrators to achieve both objectives.

Customers deploy the Dell orchestrator proxy server using Helm charts in a customer-provided Kubernetes cluster. Upon successful deployment, the Dell orchestrator proxy is onboarded to a SaaS Dell Automation Platform orchestrator.

Create client identity

Steps

1. Open a new web browser tab and go to the Dell Automation Platform portal URL.
2. Log in with administrator credentials and your updated password.
3. In the navigation pane, click **Settings** > **Identity Management** > **Clients**.
4. To create a client, click **Add**.
5. In the **Create Client** dialog:

   NOTE: Do not confuse the *Client Name* with the *Client ID*. The Client ID is a UUID string that is required for client onboarding.

   a. Enter a **Name**, for example **`dpaadmin`**.
   b. Under **Team**, select **Administrator**.
   c. Ensure the **Client enabled** slider is switched to `Enable`.
   d. In the **Client Secret** field, click the **Generate** button. Copy and save the name and secret for future use.

CAUTION: Do not use the default Client ID. Doing so results in failure to retrieve the access token.

e. Click **Close** to create the client.

Deploying the SaaS Dell orchestrator proxy with Kubernetes

The Dell orchestrator proxy can be installed in various virtualization environments, depending on each existing customer networking infrastructure. This section provides an example of installing Dell orchestrator proxy on a Kubernetes cluster.

Deploy the Dell orchestrator proxy into Kubernetes with Helm

Follow this procedure to deploy the Dell orchestrator proxy into Kubernetes with Helm.

Prerequisites

Before deploying Dell orchestrator proxy using Helm, ensure that you have the following tools installed and configured:

- Helm v3+
- kubectl, with access to the target Kubernetes cluster
- Required values for secrets and IPs, which are detailed below

Steps

1. To pull the Helm chart from Amazon ECR, run the following command:

```
helm pull oci://public.ecr.aws/dell/oxy/orchestrator-proxy --version 1.0.0-0-11
```

   This command downloads `orchestrator-proxy-1.0.0-0-11.tgz` to your local machine.

2. If the Kubernetes namespace `oxy` does not exist, run `sudokubectl create namespace oxy`.
3. To install the Oxy service with Helm, run the following command: `sudo helm install`.
   The following is an example; you must use the values applicable to your environment, as explained in the table below.

```
sudo helm install \
--set global.dockerSecret.dockerconfigjson=eyJhdXRocyI6e319 \
--set global.oxyServer.name=oxy_prod \
--set global.oxyServer.fqdn=10.229.17.91 \
--set global.dspo.ip=10.229.17.90 \
--set global.dspo.fqdn=mtls.edge.internal.use.only  \
--set global.dspo.secrets.oxyCloud.clientID= dbd8418c-b9bc-40f3-84a4-d8279117b7e2\
--set global.dspo.secrets.oxyCloud.clientSecret=@UC@2N@@@4@Y@p5on7ez@@v@ \
--set
global.dspo.secrets.oxyCloud.tlsCaCrt=LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSUI0akND
QVltZ0F3SUJBZ0lVWXcyaC83NjYwRTh0eG1PR0ZKNjNIWDJFRHNJd0NnWUlLb1pJemowRUF3SXcKTHpFU01CQU
dBMVVFQ2hNSlJFVk1UQzFGUkVkRk1Sa3dGd1lEVlFRREV4QmxieTVsWkdkbExtUmxiR3d1WTI5dApNQ0FYRFRJ
MU1EVXhOakEyTkRRME9Gb1lEekl3TlRVd05UQTVNRFkwTlRFNFdqQXZNUkl3RUFZRFZRUUtFd2xFClJVeE1MVV
ZFUjBVeEdUQVhCZ05WQkFNVEVHVnZMbVZrWjJVdVpHVnNiQzVqYjIwd1dUQVRCZ2NxaGtqT1BRSUIKQmdncWhr
ak9QUU1CQndOQ0FBVDg4MDJOenUwYUpCMlNDbEFLMlhKN08wNHFhaFVHMUpvb0VnUG9lYUt4NkExdgpMY0Y2Zl
dzMTZSS2VDbVlxRGl6YnFSeE90VVRmQXRHaytjYTR4K09IbzRHQU1INHdEZ1lEVlIwUEFRSC9CQVFECkFnRUdN
QThHQTFVZEV3RUIvd1FGTUFNQkFmOHdIUVlEVlIwT0JCWUVGUGMvNkhSTmpyUUFXbHQ2cjB4M093dmsKVVF0Zk
1COEdBMVVkSXdRWU1CYUFGUGMvNkhSTmpyUUFXbHQ2cjB4M093dmtVUXRmTUJzR0ExVWRFUVFVTUJLQwpFR1Z2
TG1Wa1oyVXVaR1ZzYkM1amIyMHdDZ1lJS29aSXpqMEVBd0lEUndBd1JBSWdMSmZZWUJlbHYwUTJJblgxCkZBMW
Y4M2FPc3pqdVVGMXdDdU13V1FtZTRTWUNJRm0yQWxNVHhMNDFVWkV2ZjVJR0pNUjdwdGFNMktRYkJwd0kKYjBM
dTBSUzgKLS0tLS1FTkQgQ0VSVElGSUNBVEUtLS0tLQ==  \
-n oxy orchestrator-proxy orchestrator-proxy-1.0.0-0-11.tgz
```

4. Use the information in the following table to obtain parameter descriptions:

Table 3. Parameter descriptions

Parameter	Description	How to obtain
`global.dockerSecret.dockerconfigjson`	Base64-encoded `.dockerconfigjson` secret	`oci://public.ecr.aws/dell/oxy/orchestrator-proxy` is Public, so enter `eyJhdXRocyI6e319`. This is encoded `'{"auths":{}}'`.
`global.oxyServer.name`	The name identifier for this Oxy instance	Custom name, such as `oxy_prod`.
`global.oxyServer.fqdn`	Internal IP or FQDN of the Oxy service	Assigned by the cluster or infrastructure team.
`global.dspo.ip`	IP address of the DSPO service	Provided by the DSPO or infrastructure team.
`global.dspo.fqdn`	Internal FQDN of DSPO	For on-premises deployments, use `mtls.edge.internal.use.only`.
`global.dspo.secrets.oxyCloud.clientID`	`client ID from DSPO`	Provided by the DSPO admininistrator. Generally, run `kubectl get secret -n hzp keycloak-secret -o yaml` in DSPO, and base64 decode the client: `delco-client-name`. Customers can generate this parameter in the portal client. The role is Orchestrator Administrator. The *CLIENT_ID* is the ID in the API response.
`global.dspo.secrets.oxyCloud.clientSecret`	`client secret from DSPO`	Provided by the DSPO admininistrator. Generally, run `kubectl get secret -n hzp keycloak-secret -o yaml` in DSPO and base64 decode the client `delco-client-pwd`. Customers can generate this parameter by doing the following: a. Click **Identity Management** > **Clients** and select the client. b. In the client, click **More** > **Edit more** c. Click **Generate**.
`global.dspo.secrets.oxyCloud.tlsCaCrt`	TLS CA certificate used for mutual TLS	Provide the TLS certificate from DSPO. If the TLS certificate is not required (for example, in SaaS deployments), enter `NA`. If the TLS value is `NA`, then MTLS will be treated as `NA` too. If the parameter is left blank and the DSPO HOST is connected, the pipeline will fetch TLS secrets from DSPO HOST. **NOTE:** Some kubernetes versions supports `NA`. You can use `TkEK`,

Table 3. Parameter descriptions (continued)

Parameter	Description	How to obtain

		which is the base64 encoded version of `NA`.
		`ssh <orchestrator IP>`
		Ngnix:
		`kubectl get secret -n hzp tls-credential -o jsonpath="{.data.cacert}"; echo`

5. After the installation completes, verify the deployment.
   a. Run the following command to verify the pods and services: `kubectl get pods -n oxy`.
   b. Ensure that all components are in `Running` or `Completed` state.
   c. On the Dell orchestrator, click **Administration** > **Proxy Servers** > and check that the oxy servers are listed.

Next steps

If pods crash or stay in `Pending` state, check the following:

- Namespace `oxy` exists.
- Secrets are correctly passed and valid.
- FQDNs are reachable inside the cluster.

Deploying the SaaS Dell orchestrator proxy with VMware vCenter or ESXi

The Dell orchestrator proxy protects on-premises customer devices from direct exposure to the Internet while connecting to the SaaS Dell Automation Platform orchestrator. Identities are protected as the proxy routes traffic between on-premises devices and the Dell-hosted SaaS orchestrator.

NOTE: Dell Technologies expects customers to install their VMware vCenter and ESXi environments, and ensure compliance with the requirements in this document, before the Dell orchestrator proxy deployment. Customers who want assistance to perform those tasks should contact their Dell sales representative.

CAUTION: These steps are for SaaS deployments only and do not apply for on-premises deployments.

Overview

In SaaS environments, Dell Automation Platform administrators may require the abilities to:

- Use the Catalog on a Dell Automation Platform orchestrator hosted either by Dell or in a public cloud to deploy blueprints.
- Deploy and manage assets that are in their organization's private data center securely, without exposing their devices directly to the Internet.

The Dell orchestrator proxy enables administrators to achieve both objectives.

Customers deploy the Dell orchestrator proxy server using an OVA file in a customer-provided ESXi cluster managed by VMware vCenter. Upon successful deployment, the Dell orchestrator proxy is onboarded to a SaaS Dell Automation Platform orchestrator.

Proxy prerequisites and limitations

Note the following SaaS orchestrator and portal prerequisites and limitations when deploying the Dell orchestrator proxy.

The customer supplies the ESXi cluster and VMware vCenter environment.

The following limitations apply:

NIC	One NIC only
VLAN	No configured VLAN
IPv6	No IPv6 support

CAUTION: When taking a snapshot of the Dell Automation Platform virtual appliance, do not choose the *Include virtual machine's memory* option. Doing so causes Dell Automation Platform to fail to fully start.

Firewall policy

The supplied network should allow traffic for the following ports:

Table 4. Firewall policy requirements

Port number	Traffic types	Policy	Description
80	Incoming	Allow	Allow external access to the host's HTTP service.
443	Incoming	Allow	Allow external access to the host's HTTPS service.

Download required software files

About this task

This procedure describes how to download the software files required for the Dell orchestrator proxy from the Dell support site.

To download the software that is required for the Dell orchestrator proxy:

Steps

1. Open a web browser and browse to https://dell.com/support.
2. To sign in to your Dell account, click **Sign In** at the upper right corner of the screen.
3. Enter `dell orchestrator proxy` in the search bar, and select **Dell Automation Platform** in the search results.
4. Select **Drivers & Downloads**.
5. Download the Dell orchestrator proxy OVA file.
6. After you have downloaded the ZIP file, extract the file into a folder. The extracted files are uploaded later.

Deploy the Dell orchestrator proxy OVA using the VMware vSphere Client or the VMware Host Client on an ESXi host

Follow this procedure to deploy the Dell orchestrator proxy OVA using the VMware vSphere Client or the VMware Host Client on an ESXi host.

Prerequisites

Ensure that you have a vCenter Server management host with `administrator` credentials.

Steps

1. Log in to VMware vSphere Web Client for the management vCenter using `administrator` credentials.
2. Copy the Dell orchestrator proxy OVA file locally, or a URL to the file.
3. Select **Inventory** from the menu.
4. Right-click the cluster or host to which you want to deploy the Dell orchestrator proxy and select **Deploy OVF Template**. The **Deploy OVF Template** wizard opens.
5. Click **Deploy OVF Template** to deploy the VM onto vCenter.
   a. Provide the copied file or URL.

b. Provide a name and folder.
c. Select compute resources for the VM, and select **Automatically power on deployed VM**.
d. Review the selection details.
e. Select storage for the VM.
f. Select networks, and enter the following information:
   - Public Network—The management network of the vCenter or ESXi, which can reach the public network (for example, the SaaS orchestrator).
   - Private Network—Any other network that is different from the Public Network.

6. On the **Customize template** page, verify or enter the information in the following sections:
   a. In the **VM Management Network Configuration** and **DNS Configuration** pages, configure the networking information.

      NOTE: In the *DHCP enabled* scenario, the DHCP server must be able to assign the following properties:
      - VM IP address
      - Netmask
      - Default gateway
      - DNS server IP address

      If the DHCP server cannot assign any of these properties, they must be provided by static binding, which overrides the DHCP settings.

      NOTE: In the *Static IP binding* scenario, there is no DHCP server. All properties are configured with static bindings.

      The following table describes valid entries for the *DHCP enabled* and *Static IP binding* scenarios. Dell Technologies highly recommends Static IP binding.

      NOTE: Multiple DNS IP addresses must be separated by spaces.

      Table 5. DHCP scenarios

Scenario	VM IP address	Netmask	Gateway	Primary DNS
DHCP enabled	Empty	Empty	Empty	Empty
Static IP binding	*<VM IP address>*	*<netmask>*	*<default gateway>*	*<valid DNS IP address>*

   b. In the **Proxy Onboarding Configuration** page, enter the required information.

      NOTE: `Proxy Server Name` is an optional field to customize the name displayed on the SaaS orchestrator UI. If this field is left blank, a default name is used . For example, `proxy_<vm_machine_id>`.

      For more information, see the *FQDNs for OVA appliance deployment* section in the *Dell Automation Platform Pre-deployment Guide*.

7. Verify your selections on the **Ready to complete** page, and click **Finish** to start the deployment of the OVA.
8. Wait for the VM to power on, and then monitor the progress.

   You can follow the progress of the deployment in the **Recent Tasks** window.

   In the console screen, when the configuration is complete, the message `Installation has completed successfully` displays.

Results

- Once the Dell orchestrator proxy virtual appliance has been deployed successfully, a message displays to indicate that the proxy is successfully deployed and is pinging the SaaS Dell Automation Platform orchestrator.
- The virtual machine can be accessed by launching it from the vSphere web console.
- The Dell orchestrator proxy should leverage an encrypted communication with the SaaS Dell Automation Platform orchestrator.

- The SSH protocol to the Dell orchestrator proxy appliance should be disabled. The proxy can be managed only through the SaaS Dell Automation Platform orchestrator.

Next steps

After onboarding is complete, verify that the proxy connection has appeared in the Dell Automation Platform orchestrator UI.

1. Log in to Dell Automation Platform orchestrator.
2. Go to **Administration** > **Proxy Servers.**
3. Verify that the Dell orchestrator proxy appears in the list.

Deploy the Dell orchestrator proxy OVA on ESXi

Follow this procedure to deploy the Dell orchestrator proxy on ESXi.

Prerequisites

Ensure that you have an ESXi host with `root` credentials.

About this task

The procedure to deploy Dell orchestrator proxy on ESXi is generally the same as for deploying on vCenter. See Deploy the Dell orchestrator proxy OVA using the VMware vSphere Client or the VMware Host Client on an ESXi host for the specific settings for VM, DNS, and application configuration.

Steps

1. Log in to the ESXi host using `root` credentials.
2. Copy the Dell Automation Platform OVA file locally.
3. Click **Create/Register VM** to deploy the VM onto ESXi.
   a. Enter the VM name.
   b. Select the downloaded OVA file.
   c. Select storage for the VM.
   d. Select the network for the VM.
4. In the **Additional settings** page, enter the following information:
   a. **Virtual Machine Network Configuration** and **DNS Configuration**
   b. **Application Configuration**
5. Verify your selections on the **Ready to complete** page, and click **Finish** to start the deployment of the OVA.
6. Wait for the VM to power on, and then monitor the progress.

Results

- Once the Dell orchestrator proxy virtual appliance has been deployed successfully, a message displays to indicate that the proxy is successfully deployed and is pinging the SaaS Dell Automation Platform orchestrator.
- The virtual machine can be accessed by launching it from the vSphere web console.
- The Dell orchestrator proxy should leverage an encrypted communication with the SaaS Dell Automation Platform orchestrator.
- The SSH protocol to the Dell orchestrator proxy appliance should be disabled. The proxy can be managed only through the SaaS Dell Automation Platform orchestrator.

Next steps

After onboarding is complete, verify that the proxy connection has appeared in the Dell Automation Platform orchestrator UI.

1. Log in to Dell Automation Platform orchestrator.
2. Go to **Administration** > **Proxy Servers.**
3. Verify that the Dell orchestrator proxy appears in the list.

4

Deploying the on-premises Dell Automation Platform

This section covers deploying the on-premises Dell Automation Platform with Kubernetes or VMware vCenter or ESXi.

Topics:

- Deploying the on-premises Dell Automation Platform with Kubernetes
- Deploying the on-premises Dell Automation Platform with VMware vCenter or ESXi
- Initializing the Dell Automation Platform portal and orchestrator for on-premises deployments

Deploying the on-premises Dell Automation Platform with Kubernetes

Dell Automation Platform can be installed in various on-premises virtualization environments, depending on each customer's existing networking infrastructure. This chapter provides an example of scenarios in which Dell Automation Platform is installed on either a single-node or three-node Kubernetes cluster.

NOTE: Users who choose to deploy into an existing Kubernetes environment are required to deploy into a cluster with the ability to fulfill requests for PersistentVolumeClaims.

This section explains the process for install the Dell Automation Platform into a Kubernetes environment. It also describes a noninteractive installation mode to support automation use cases.

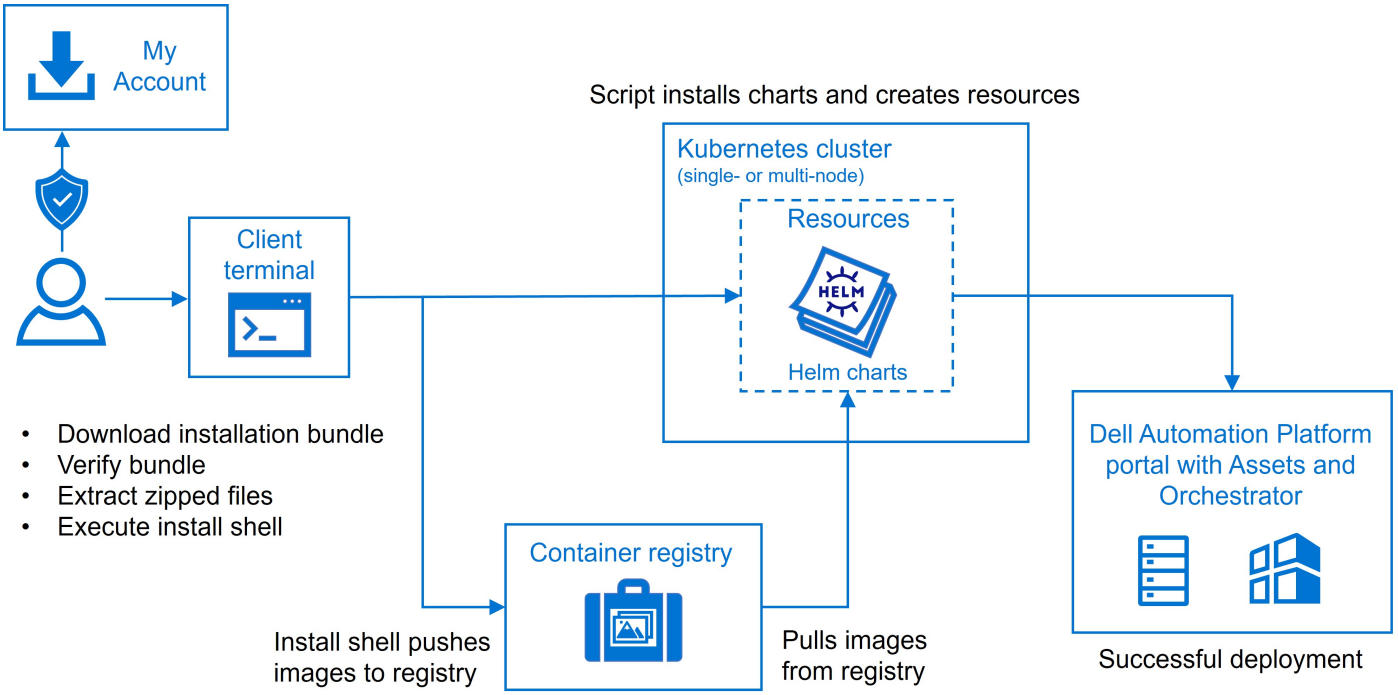

Figure 1. Dell Automation Plafform on-premises deployment overview

Download and prepare the Dell Automation Platform installation bundle

Prerequisites

CAUTION: Dell Technologies recommends that users implement backup and restore procedures to a Dell Automation Platform deployment. For more information and an example backup and restore workflow, see Backup and restore.

If the Kubernetes cluster has been deployed on a virtual machine, Dell Technologies recommends taking a VM snapshot before running the deployment script. Doing so ensures a recovery method for a failed deployment.

Steps

1. Download the `DellAutomationPlatform_v<version>.zip` installation bundle from Dell My Account or the Dell Automation Platform support site and upload it to the host used for the Dell Automation Platform installation.
2. Extract the `DellAutomationPlatform_v<version>.zip` installation bundle:

   ```
   unzip DellAutomationPlatform_v<version>.zip
   ```

   The installation ZIP bundle contains three files:

DellAutomationPlatform_v*<version>*-*<build>*.zip	Supporting scripts, manifests, and images
DellAutomationPlatform_v*<version>*-*<build>*.zip.signed.bin	Signature file for bundle verification
DellAutomationPlatformHZP_30_public.txt	Dell public key for signature verification

3. Use the `signed.bin` and public key file to verify the extracted bundle:

   ```
   openssl dgst -sha384 -verify DellAutomationPlatformHZP_30_public.txt \
   -signature DellAutomationPlatform_v<version>-<build>.zip.signed.bin \
   DellAutomationPlatform_v<version>-<build>.zip
   ```

4. Extract the `DellAutomationPlatform_v<version>-<build>.zip` file into a new directory:

   ```
   unzip DellAutomationPlatform_v<version>-<build>.zip -d <directory_name>/
   ```

5. Change to the new directory and add executable permission to the deployment script:

   ```
   chmod +x ./install-upgrade.sh
   ./install-upgrade.sh
   ```

Run the deployment script

Prerequisites

Ensure that all components are in compliance with the *Dell Automation Platform 1.x Support Matrix* and the *Dell Automation Platform 1.x Pre-deployment Guide* listed versions of Kubernetes, Docker, and Helm.

The deployment script has several parameters for deploying in an interactive or noninteractive, silent method. The deployment script must be run with a user that has `sudo` privileges, or as the root user.

To verify the user is properly configured, the following commands should return without errors:

```
sudo kubectl get nodes
sudo helm list
sudo docker ps
```

Also, verify that Docker can log in to the registry:

```
sudo docker login <registryURL>
```

By default, the `install-upgrade.sh` script may be run without any parameters, which starts the deployment in an interactive method, prompting the user for the following inputs:

Parameter	*Description*	Example
`EO_HOST`	*<Fully Qualified Domain Name for Dell Automation Platform orchestrator>*	`orchestrator.host.local.edge`
`IMAGE_REG_URL`	*<registry url>/<repository>*	`registry.local.edge/project`
`IMAGE_REG_USERNAME`	*<registry login username>*	`administrator`
`IMAGE_REG_PASSWORD`	*<registry login password>*	`Password123!`
`REGISTRY_CERT_FILE_PATH`	*<registry certificate path>*	`/home/username/registry.crt`
`NAMESPACE`	*<installation namespace>*	`hzp`
`PORTAL_NAMESPACE`	*<portal installation namespace>*	`Portal`
`PORTAL_COOKIE_DOMAIN`	*<portal installation cookie domain>*	`host.local.edge`
`PORTAL_INGRESS_CLASS_NAME`	*<portal installation ingress class name>*	`haproxy`
`PORTAL_HOST`	*<portal Fully Qualified Domain Name>*	`portal.host.local.edge`
`ORG_NAME`	*<organization name>*	`Someorg`
`ORG_DESC`	*<organization description>*	`Some description`
`FIRST_NAME`	*<administrator first name>*	`Firstname`
`LAST_NAME`	*<administrator last name>*	`Lastname`
`USERNAME`	*<administrator username>*	`Administrator`
`EMAIL`	*<administrator email address>*	`administrator@local.edge`

NOTE: The `portal_cookie_domain` is the common part of the FQDN's used by the `Portal_Host` and the `EO_Host`. This allows the cookie of the session be used across both portal and orchestrator calls from a web browser.

Deployment script syntax and parameters:

```
sudo ./install-upgrade.sh [parameter1=value] [parameter2=value] [and so on...]
```

Required parameters for use in a noninteractive install:

Parameter	Description
`EO_HOST`	*<Fully Qualified Domain Name for Dell Automation Platform orchestrator>*
`IMAGE_REG_URL`	*<registry url>/<repository>*
`IMAGE_REG_USERNAME`	*<registry login username>*
`IMAGE_REG_PASSWORD`	*<registry login password>*

Optional parameters:

Parameter	Description
SKIP_IMAGES_LOADER	*<true/false>*
RETRY_ON_FAILURE	*<true/false>* **CAUTION: This parameter should never be set to *true* for this release.**
RETRY_ON_FAILURE_COUNT_MAX	*<max number of retries on failure>*
IMAGE_LOADER_THREAD_COUNT	*<number of threads in eo-images-loader>*—how many threads used in `eo-images-loader` to load and push images. The default is 5. The maximum threads that are allowed are 20.

NOTE: On the initial deployment, if Dell Automation Platform orchestrator images have not yet been pushed to the registry, `SKIP_IMAGES_LOADER` should not be set to true.

NOTE: Before running the deployment script, manually confirm that all prerequisites are satisfied.

About this task

The deployment script performs the following high-level steps to prepare the environment for the Dell Automation Platform orchestrator installation:

- Performs preflight checks to ensure that versions meet the requirements that are defined in the *Dell Automation Platform 1.x Pre-deployment Guide*.
- Loads container images into the registry.
- Deploys the Dell Automation Platform orchestrator.

The deployment script can be run in either an interactive or noninteractive, parameterized fashion.

Steps

1. Run the deployment script in one of the following modes:
   - As an interactive install:

     ```
     sudo ./install-upgrade.sh
     ```

   - As a noninteractive install:

     ```
     sudo ./install-upgrade.sh EO_HOST=orchestrator.host.local.edge
     IMAGE_REG_URL=registry.local.edge/project IMAGE_REG_USERNAME=administrator
     IMAGE_REG_PASSWORD=Password123! REGISTRY_CERT_FILE_PATH=/home/username/registry.crt
     NAMESPACE=hzp PORTAL_NAMESPACE=Portal PORTAL_COOKIE_DOMAIN=host.local.edge
     PORTAL_INGRESS_CLASS_NAME=haproxy PORTAL_HOST=portal.host.local.edge
     ORG_NAME=Someorg ORG_DESC=Some description FIRST_NAME=Firstname LAST_NAME=Lastname
     USERNAME=Administrator EMAIL=administrator@local.edge
     ```

2. When prompted, enter the namespace for the installation.
3. In the resulting output code, confirm that preflight checks have all passed, as shown in the following example:

   ```
   ...
   Operation: INSTALL
   Validating input parameters
   Performing pre-requisite checks...
   Checksum                  : Passed
   Docker                    : Passed
   jq                        : Passed
   Kubernetes version Check  : Passed
   Helm version Check        : Passed
   All pre-requisite checks have passed successfully.
   ```

4. If an interactive installation was run, the script prompts the user to provide inputs.

   ```
   Enter EO host for Dell Automation Platform Orchestrator:
   orchestrator.host.local.edge
   Enter container registry url (in the format of <registry>/<repository>, ex: docker.io/
   dell):
   ```

```
registry.local.edge/project
Enter container registry username:
admin
Enter container registry password:

Enter container registry certificate path (ex: ./dockerca.crt):
/usr/local/share/ca-certificates/harbor.crt
Enter Cookie Domain for Portal:
host.local.edge
Enter Ingress Class Name for Portal:
haproxy
Enter host for Portal:
portal.host.local.edge
```

5. The deployment script launches an image loader, which logs in to the remote registry and pushes the individual Dell Automation Platform container images to the registry.

```
Creating the namespace hzp
namespace/hzp created
namespace/hzp labeled
Creating the namespace portal
namespace/portal created
namespace/portal labeled
Creating patch files with the user inputs
Saving configmap at /home/eoadmin/neo/.resources/configmap.yaml
Saved ConfigMap successfully
Creating registry secret eo-registry-secret in namespace hzp
secret/eo-registry-secret created
Creating docker config json secret docker-secret in namespace hzp
secret/docker-secret created
Creating docker config json secret docker-secret in namespace portal
secret/docker-secret created
Preparation is complete!
Login Succeeded
Loading EO images loader image locally
Loaded image: hzp/eo-images-loader:1.0.0.0
Starting EO Images Loader
Waiting for EO image loader to start reading updates
2025/07/16 21:45:31 login to registry dapharbor.sre.edge/dap28d9ffc
2025/07/16 21:45:32 loading manifest from output/manifest.yaml
2025/07/16 21:45:32 Pushing 119 images
2025/07/16 21:45:32 Attempt [1/5] to parse component [busybox:1.37.0]
2025/07/16 21:45:32 Attempt [1/5] to parse component [hzp/hzp-util:0.1.3]
2025/07/16 21:45:32 Attempt [1/5] to parse component [hzp/hzp-fido-owner-
reseller:1.0.0.0-4328b1c-91c75677]
2025/07/16 21:45:32 Attempt [1/5] to parse component [hzp/fido-onboarding-
svc:1.0.0.0-9077bbe-6d700e92]
2025/07/16 21:45:32 Attempt [1/5] to parse component [hzp/fru-cru-
svc:3.2.0.0-0f3f3d7-315113ef]
2025/07/16 21:45:33 pushed image [busybox:1.37.0], progress: 1/119
...
```

6. Once the image upload process is complete, the deployment script installs Dell Automation Platform orchestrator. In interactive mode, the script opens the `helm values` file or manual editing. By default, no editing is necessary for Dell Automation Platform orchestrator. Save and exit the `helm values` file by typing **`:wq`** in the text editor.

```
Getting image loader logs
EO Image Loader completed
Deleting EO image loader
EO Images loader uninstalled
Opening Helm values file for manual edits
```

7. The deployment script continues to install Dell Automation Platform orchestrator and monitors the progress of the deployment.

```
eo-management:
  nginx_gw:
    nginx:
      sslProtocols: "TLSv1.2 TLSv1.3"
  seaweedfs:
```

```
    volume:
      data:
        size: 100Gi
global:
  imagePullSecrets:
  - docker-secret
  imageRegistry: registry.local.edge/container
  ingress:
    exposeEOServices: true
    fqdn: <FQDN>
  opensearch:
    resources:
      storageSize: 45Gi
  s3:
    enableHA: false
  storageClass: null
Creating configmap.
configmap/delco-config created
Chart exists. Skip unarchive...
NEO Installation has started
OperationType: INSTALL OperationStatus: IN_PROGRESS ChartKey: NEO
OperationType: INSTALL OperationStatus: COMPLETED ChartKey: NEO
OperationType: POST_INSTALL OperationStatus: IN_PROGRESS
Getting delco service logs
Deleting delco-svc deployment
OperationType: POST_INSTALL OperationStatus: COMPLETED
native-edge-orchestrator has deployed successfully
Saving logs to files
Cleaning up the execution log file
Taking log dumps
Zipping log files
Cleaning the files
ConfigMap 'neo-custom-values' created successfully with the key 'neo-custom-
values.yaml'.
Installation has completed successfully
```

8. Once the installation has been completed successfully, the default credentials are printed to the console and the script exits. The output of a successful installation provides the Dell Automation Platform orchestrator URL, default username, and a default (one-time use) password.

```
...
Installation has completed successfully
------------------------------------------------------------
username: administrator
password: <one-time password>
portal url: <FQDN>
orchestrator url: <FQDN>
------------------------------------------------------------
```

Deploying the on-premises Dell Automation Platform with VMware vCenter or ESXi

Dell Automation Platform can be installed in various on-premises virtualization environments, depending on each customer's existing networking infrastructure. Dell Automation Platform can be installed in an on-premises VMware virtualization environment. This section provides an example of scenarios in which Dell Automation Platform is installed on ESXi host using VMware vSphere client or VMware host client.

NOTE: Prior to starting deployment, ensure that the environment meets the prerequisites and limitations that are described in the *Dell Automation Platform 1.x Pre-deployment Guide* available in the Dell support site.

Download required software files

This procedure describes how to download the software files required for Dell Automation Platform from Dell Support.

Prerequisites

To download the software that is required for Dell Automation Platform:

Steps

1. Open a web browser and browse to the Dell Support page.
2. To sign in to your Dell account, click **Sign In** at the upper right corner of the screen.
3. Enter **`dell automation platform`** in the search bar, and select **Dell Automation Platform** in the search results.
4. Select **Drivers & Downloads**.
5. Download the file Dell Automation Platform OVA file. The filename is similar to: `Dell_Automation_Platform_v1.0.0.0-xxxxxxxx _OVF10.x86_64-15.6-signed.ova`.

Deploy the Dell Automation Platform OVA on vCenter

Follow this procedure to deploy Dell Automation Platform OVA using the VMware vSphere Client.

Prerequisites

Ensure that you have a vCenter Server management host with `administrator` credentials.

Steps

1. Log in to VMware vSphere Web Client for the management vCenter Server using `administrator` credentials.
2. Copy the Dell Automation Platform OVA file locally, or a URL to the file.
3. Click **Deploy OVF Template** to deploy the VM onto vCenter.
   a. Provide the copied file or URL.
   b. Provide a name and folder.
   c. Select compute resources for the VM, and select **Automatically power on deployed VM**.
   d. Review the selection details.
   e. Select storage for the VM.
4. On the **Customize template** page, verify or enter the information in the following sections:
   a. In the **Network Configuration** and **DNS Configuration** pages, configure the networking information.

      NOTE: In the *DHCP enabled* scenario, the DHCP server must be able to assign the following properties:
      - VM IP address
      - Netmask
      - Default gateway
      - DNS server IP address

      If the DHCP server cannot assign any of these properties, they must be provided by static binding, which overrides the DHCP settings.

      NOTE: In the *Static IP binding* scenario, there is no DHCP server. All properties are configured with static bindings.

      The following table describes valid entries for *DHCP enabled* and *Static IP binding* scenarios.

      NOTE: Multiple DNS IP addresses must be separated by spaces.

Table 6. DHCP scenarios

Scenario	VM IP address	Netmask	Gateway	Primary DNS
DHCP enabled	Empty	Empty	Empty	Empty
Static IP binding	*<VM IP address>*	*<netmask>*	*<default gateway>*	*<valid DNS IP address>*

b. In the **Dell Automation Platform Application Configuration** page, enter the required information as in the following example.

Table 7. Example single-node DNS schema

Function	IP address	DNS	Note
Host	192.168.0.50	host.domain.edge	Server that hosts the deployment
Portal UI/API	192.168.0.50	portal.host.mydomain.edge	UI/API services derived from load balancer IP address
Orchestrator UI/API	192.168.0.50	orchestrator.host.mydomain.edge	UI/API services derived from load balancer IP address
mtls	192.168.0.50	mtls-orchestrator.host.mydomain.edge	Used by Dell Automation Platform infrastructure for mtls connection exposed from load balancer IP address
mtls-recovery	192.168.0.50	mtls-recovery-orchestrator.host.mydomain.edge	Used by Dell Automation Platform infrastructure for mtls recovery connection exposed from load balancer IP address

The **Portal Cookie Domain** is the base domain of two FQDNs. For example, if two FQDNs are `c1.portal.dap.rack001.local` and `c1.orchestrator.dap.rack001.local`, then the cookie domain is `rack001.local`.

NOTE: By default the FQDN validation performs the following checks:

- The orchestrator FQDN, portal FQDN, and portal cookie domain should be provided.
- The portal cookie domain should match both orchestrator FQDN and portal FQDN.
- With a provided DNS server, the orchestrator FQDN and portal FQDN should both resolve to the VM IP address.

NOTE: In certain scenarios, the FQDNs do not resolve to the VM IP address:

- A centralized Nginx server takes over all the requests and dispatches the packages to the corresponding Dell Automation Platform VM. In this case, FQDNs should be resolved to the Nginx server IP address instead of the Dell Automation Platform VM IP address.

In order to ensure that the Dell Automation Platform installation is successful, the FQDN validations must be skipped.

CAUTION: Selecting the Skip Domain Name Check checkbox bypasses all FQDN checks, which could cause unpredictable errors. Users are responsible for ensuring that their configurations are correct.

For more information, see the *FQDNs for OVA appliance deployment* section in the *Dell Automation Platform Pre-deployment Guide*.

c. In the **User information** page, enter the following information:

- Organization Name
- Organization Description
- First Name
- Last Name
- Username
- Email

After deployment, the administrator uses the Username and Email information to log in.

5. Verify your selections on the **Ready to complete** page, and click **Finish** to start the deployment of the OVA.
6. Wait for the VM to power on, and then monitor the progress.

   You can follow the progress of the deployment in the **Recent Tasks** window.

   In the console screen, when the configuration is complete, the message `Installation has completed successfully` displays. Directly following this message is the **Default login credentials to access the Dell Automation Platform orchestrator** section, which lists the username, password, and URL you must provide to log in to Dell Automation Platform orchestrator later.

   NOTE: Record the username, unique password, and URL that are displayed in the next step. You cannot proceed without these items.

Deploy the Dell Automation Platform OVA using the VMware Host Client on a standalone ESXi host

Follow this procedure to deploy Dell Automation Platform using the VMware Host Client on a standalone ESXi host.

Prerequisites

Ensure that you have an ESXi host with `root` credentials.

About this task

The procedure to deploy Dell Automation Platform on ESXi is generally the same as for deploying using the VMware vSphere Client. See Deploy the Dell Automation Platform OVA on vCenter for the specific settings for VM, DNS, and application configuration, and monitoring the deployment progress.

Steps

1. Log in to the ESXi host using `root` credentials.
2. Copy the Dell Automation Platform OVA file locally.
3. Click **Create/Register VM** to deploy the VM onto ESXi.
   a. Enter the VM name.
   b. Select the downloaded OVA file.
   c. Select storage for the VM.
   d. Select the network for the VM.
4. In the **Additional settings** page, enter the following information:
   a. **Virtual Machine Network Configuration** and **DNS Configuration**
   b. **Application Configuration**
5. Verify your selections on the **Ready to complete** page, and click **Finish** to start the deployment of the OVA.
6. Wait for the VM to power on, and then monitor the progress.

Optional vTPM support for the Dell Automation Platform orchestrator OVA

A Virtual Trusted Platform Module (vTPM) is a software-based representation of a physical Trusted Platform Module (TPM) 2.0 chip. With vTPM, each Dell Automation Platform orchestrator OVA-based appliance can have its own unique and isolated TPM to help secure sensitive information and ensure system integrity.

Prerequisites

vTPM needs a key provider that is configured in vCenter to enable VM encryption. If you have not configured it yet, see the following VMware vSphere documentation:

- Configure a vSphere Native Key Provider
- Add a Standard Key Provider Using the vSphere Client

About this task

If the key provider is configured properly, you can add a vTPM virtual device by following these steps:

Steps

1. Right-click the Dell Automation Platform orchestrator virtual machine in the inventory and select **Edit Settings**.
2. In the **Edit Settings** dialog:
   a. Click **Add New Device**.
   b. Select **Trusted Platform Module**.
   c. Click **OK**.

   The virtual machine **Summary** tab now includes the Virtual Trusted Platform Module in the **Virtual Hardware** pane.
3. Power on the Dell Automation Platform orchestrator VM.

Accept the EULA, upload the license, and set the encryption key escrow

To complete the initial configuration of the Dell Automation Platform orchestrator, users must accept the EULA, upload the license, and set the encryption key escrow by entering a secure passphrase that meets the specified policy requirements.

Steps

1. Open a web browser tab and go to the URL provided in the **Default login credentials to access Dell Automation Platform orchestrator** section that appears on the console screen. The Dell Automation Platform orchestrator login page appears.
2. Enter the administrator credentials from the **Default login credentials to access Dell Automation Platform orchestrator** console screen and click **Log In**.
3. You are prompted to enter a **New Password** and then to **Confirm password**. The password policy appears on the screen with password convention requirements. Click **Submit**.
4. You must update your user profile to activate your account. Enter the email, first name, and last name affiliated with the administrator account. Click **Submit**.
5. When the **Initial Configuration** wizard opens, read the EULA, scroll down, click **I accept this agreement**, and click **Accept**.

Next steps

Finalize the initialization as described in the following section.

Initializing the Dell Automation Platform portal and orchestrator for on-premises deployments

This section describes the first-time setup steps for your Dell Automation Platform on-premises portal and orchestrator, along with suggestions to ensure that your Dell Automation Platform account is entitled and ready for immediate usage.

There is a licensing issue with deploying the orchestrator with Dell Connectivity Services for the first time. The workaround for Dell Automation Platform version 1.0 is to deploy in the following sequence:

1. When activating the orchestrator, skip the Dell Connectivity Services configuration and choose the unconnected option. For more information, see License the orchestrator in an on-premises, unconnected environment.
2. Log in to My Account to generate the license file.
3. Upload the license file during orchestrator deployment.
4. Enable Dell Connectivity Services as part of Day-2 configuration tasks by generating an access key and PIN from the Dell Automation Platform Support site.

NOTE: This issue will be addressed in a future release.

Initialize the Dell Automation Platform for the first time

About this task

When you first log in to the Dell Automation Platform portal, you must complete a series of one-time steps to initialize the service:

Steps

1. Read the Dell Automation Platform Terms and Conditions, select the checkbox that you have read and agree to all the terms and conditions, and click **Accept**.

   After you accept the agreement, the Dell Automation Platform automatically downloads ownership vouchers from Dell MyAccount for hardware devices that support it. This enables automated and secure onboarding of these applicable devices into the Dell Automation Platform. As vouchers download to your portal, the associated devices are displayed in the **Assets** page of the portal.

   NOTE: Device vouchers might become available for download later, as new Dell hardware is ordered and shipped to your location. Agreeing to this one-time prompt allows all current and future vouchers to download automatically, enabling automatic onboarding of the associated hardware devices.

2. To activate your Dell Automation Platform orchestrator instance, navigate to your orchestrator using the main portal display or when prompted in the **Portal Assets** screen.

   When you navigate to your SaaS-based orchestrator for the first time, it is licensed from an available entitlement that is associated with your Dell Automation Platform account. If you do not have an entitlement to use the Dell Automation Platform SaaS orchestrator, you are prompted to provide a license before proceeding. Contact Dell Sales to purchase an entitlement, or contact Dell Support to identify an existing entitlement.

   If you have purchased NativeEdge devices, you have already agreed to allow voucher downloads. If your orchestrator is properly licensed, you see an alert informing you that an orchestrator administrator must enter a NativeEdge Escrow Key Passphrase before NativeEdge devices can onboard to the orchestrator. Click the link in the alert and then follow the on-screen instructions.

3. To create an **Encryption Key Escrow Passphrase**, hover over the information icon for details about complexity requirements.

   CAUTION: It is critical that you take note of the Encryption Key Escrow Passphrase now. It cannot be retrieved later, and it will be required for future restore and replacement tasks.

   NOTE: If you do not yet own NativeEdge devices or have not downloaded their associated vouchers to your Dell Automation Platform portal, you will not see the alert regarding the passphrase. However, you can still provide a passphrase if you anticipate onboarding NativeEdge devices in the future. To do so, go to your orchestrator and navigate to the **Security** page under the **Administrator** side navigation pane. Next, navigate to the **NativeEdge Escrow Passphrase** tab, and follow the on-screen instructions to create an acceptable passphrase.

On-premises licensing

For connected on-premises Dell Automation Platform installations, you must set up Remote Dell connectivity to enable automated licensing operations. For unconnected sites (locations where no Internet connection is available or allowed), follow the file-based license procedures that are outlined later. For details, see License the orchestrator in an on-premises, unconnected environment.

Access configuration requirements for deployment and Day-2 operations

To ensure entitlement visibility, licensing accuracy, and compliance with Dell's licensing policies, the following access configurations are mandatory during both initial deployment and ongoing operations.

Access key and PIN configuration for SaaS and on-premises deployments

The access key and PIN is used to configure connectivity to Dell for both SaaS and on-premises Dell Automation Platform deployments. In SaaS, it is used for Dell Support only, while on-premises deployments use it for support and licensing operations.

CAUTION: The access key and PIN used to configure Dell Connectivity Services must belong to the Customer Administrator.

NOTE: Credentials from deployment engineers or temporary users must not be used. Other methods to configure the access key and PIN are strictly prohibited.

Understanding how access keys and PINs function are important because the Dell Automation Platform orchestrator retrieves license entitlements based on the identity that is linked to the access key and PIN. If incorrect credentials are used, entitlements may be mapped to the wrong organization. This can lead to several issues, including loss of entitlement visibility for the customer, data integrity problems, and failures in compliance reporting.

For details on how to generate an access key and PIN, see Generate an access key on the Dell support site.

CAUTION: Compliance note—Any attempt to bypass this requirement violates licensing policy and may result in audit failures or support limitations.

User profile association for license visibility for SaaS deployments

Each user who requires access to license entitlements must ensure that their Dell Identity profile is correctly linked to their organization profile in the Customer Profile Data (CPD) system.

This is important because if the user's Dell Identity profile is not linked to the correct organization, they will not see any license entitlements even if they are authorized.

If a user is unable to view license entitlements, raise a service request from within Dell Automation Platform.

As a best practice, include Dell Identity profile validation as part of your user-provisioning checklist.

Hierarchy validation for on-premises connected deployments

For Dell Connectivity Services to function correctly, the administrator whose access key is used must be positioned within the hierarchy so that they have access to the required entitlements. This requirement is critical because entitlement visibility within the Dell Automation Platform is dictated by the administrator's position within the hierarchy. They must have access to entitlements that were purchased for a primary location in the hierarchy, as well as any entitlements purchased for child locations. The user does not have access to any entitlements purchased for a location that is positioned above the administrator user's location or in a separate branch in the hierarchy.

Misalignment can result in incomplete license provisioning, missed renewals, and operational blind spots. To prevent these issues, it is essential to confirm the CPD hierarchy before deployment—either by using internal tools or by contacting Dell Support. Additionally, ensure that the access key owner has full visibility into all entitlements across the customer's organization.

CAUTION: Compliance risk—Misalignment in the hierarchy can result in partial license visibility, which may affect compliance reporting and support eligibility.

Verify the installation and license Dell Automation Platform orchestrator using secure support connectivity

About this task

Perform the following steps to verify the Dell Automation Platform orchestrator installation using secure support connectivity.

Steps

1. Go to the following location from a host on the network that can access the Dell Automation Platform orchestrator: https://*<orchestrator FQDN>* . You may receive a certificate warning due to the self-signed certificate used by the orchestrator. Log in using the initial credentials output by the installation script.
2. Click **Administration** > **Support**.
3. Select the **Secure remote** tab and then click **Configure**.
4. Enter the access key and PIN used for connectivity.
5. Confirm that all initial configuration results are successful and click **Close**.
6. Confirm that the Dell Automation Platform orchestrator dashboard loads successfully.

Verify the installation and license Dell Automation Platform orchestrator in an unconnected environment

About this task

Perform the following steps to verify the Dell Automation Platform orchestrator installation in an unconnected environment. This task includes licensing of the orchestrator software.

Steps

1. Go to the following location from a host on the network that can access the Dell Automation Platform orchestrator: https://*<orchestrator FQDN>* . You may receive a certificate warning due to the self-signed certificate used by the orchestrator. Log in using the initial credentials output by the installation script.
2. Provide an updated password that meets the password complexity requirements.
3. Provide the **Email** address, **First name**, and **Last name** of the administrator account.
4. The **Initial Configuration** window launches. Review and accept the **End User License Agreement** (EULA).
5. Complete the on-screen instructions to obtain and upload a license to Dell Automation Platform orchestrator.
6. Generate a license from Dell My Account.

   Licenses for Dell Automation Platform orchestrator are obtained by using license authorization code (LAC) from Dell My Account to activate the license on the Software Licensing Central.

   Each Dell Automation Platform orchestrator entitlement is tied to several Dell Automation Platform infrastructure assets, and based off how many assets are purchased during the time of the order.

   Perform the following substeps to generate a license for Dell Automation Platform orchestrator.

   a. Log in to Dell My Account and select the appropriate Dell Automation Platform orchestrator Instance License from the **Software & Licenses** page.
   b. From the **Product Details** tab, click the **Activate Now** button to be redirected to **SOFTWARE LICENSING CENTRAL** page.
   c. In the **ACTIVATE** tab, select the checkbox next to the Dell Automation Platform Device Category and Instance licenses you want to associate with this Dell Automation Platform orchestrator. Then click **START THE ACTIVATION PROCESS**.
   d. Verify that the **COMPANY DETAILS** are correct, then click **NEXT: SELECT A MACHINE**.
   e. Select an existing machine or create a new machine if none exists. Then click **SAVE MACHINE & CONTINUE TO NEXT STEP**.
   f. In the **Quantity to Activate** column, update the number of Instance Licenses to 1 and Device Category licenses to match the number of Dell Automation Platform assets to be used. Then, under **ENTER MACHINE DETAILS**, paste in the **Dell Automation Platform orchestrator Unique Identifier** copied from the Dell Automation Platform UI. Then click **NEXT: REVIEW**.
   g. In the **REVIEW** tab, verify that the information is correct. Click the **Email to more people** to email a copy of the license file to additional recipients. Then, click **ACTIVATE >**.
   h. Download the license, or obtain the license file from the automated email.
   i. Navigate back to the Dell Automation Platform orchestrator UI and upload the license file.
7. Create an **Encryption Key Escrow** Passphrase. Hover over the information icon for details about complexity requirements.

   WARNING: It is critical to take note of the Encryption Key Escrow *Passphrase* now. It cannot be retrieved later, and it will be required for future restore and replacement tasks.

8. Confirm that all initial configuration results are successful, and click **Close**.
9. Confirm that the Dell Automation Platform orchestrator dashboard loads successfully.

License the orchestrator in an on-premises, unconnected environment

The initialization screens for the Dell Automation Platform orchestrator prompt you to connect to the license service, with the option to choose **No direct connection**. If you choose this option, you must generate license files from the Dell My Account portal and upload those files to your Dell Automation Platform installation.

Steps

1. Copy the License Identifier for your on-premises installation, using the associated link on the licensing screen.

2. Next, navigate to Dell MyAccount in a separate browser tab using the appropriate link in the same licensing screen.
3. When you are prompted, log into Dell MyAccount using your Dell username and password.
4. Locate your license within the list of entitlements or search for it within the Software Licenses tab of MyAccount.

   Licenses for Dell Automation Platform are obtained by using the license authorization code (LAC) from MyAccount to activate the license. The LAC might have also been emailed to you.
5. After locating and selecting the license entitlement to activate, click **Manage** button on the right and select **View Details**.
6. Read through the notes displayed within the **Important Information** section, and click **Activate Now**.
7. Choose all the entitlements that apply on the next screen and then click **Next**.
8. For a new installation, provide a machine name in the **Add a new machine** text field and then click **Add**.
9. On the **Details** page, enter the following information:
   a. Enter the quantity you are activating, which will be `1` for an on-premises installation.
   b. In the Locking ID section, paste the License Identifier you copied in step 1 in the **Orchestrator License Unique ID** field.
   c. Click **Next**.
10. Ensure that all of the information entered, the associated organization information, and the location of the installation are correct and then click **Activate**.

    NOTE: To deliver the license file to additional email addresses, enter them in the appropriate text field.

    The license will be delivered to the email address associated with the entitlement and any addresses provided
11. When prompted at the time of activation, download the license file within MyAccount.

    If you closed the Dell Automation Platform orchestrator without providing the license file, select the orchestrator panel within the Dell Automation Platform Portal. The panel displays your most recent view.

5

Troubleshooting deployment issues

Topics:

- Troubleshooting a Kubernetes deployment

Troubleshooting a Kubernetes deployment

This section describes common commands and troubleshooting approaches for working with Kubernetes. It is not a comprehensive list of Kubernetes troubleshooting approaches, and Dell Automation Platform administrators should possess proficiency with maintaining and troubleshooting Kubernetes clusters.

Troubleshoot deployments

About this task

The first step when troubleshooting an issue with services running on Kubernetes is to determine the health of the relevant deployments. Perform the following steps to troubleshoot an issue with a deployment.

Steps

1. Log in as `root` to the host on which Dell Automation Platform is deployed.
2. Run the `kubectl get ns` to list the relevant namespaces.

   The following is a full list of deployments for the orchestrator namespace:

```
kubectl -n hzp get deployments
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
eo-management-public-fileserver     1/1     1            1           33d
eo-management-seaweedfs-s3          1/1     1            1           33d
fido-manufacturer                   1/1     1            1           33d
fido-owner                          1/1     1            1           33d
fido-rv                             1/1     1            1           33d
fusion-api-service                  1/1     1            1           33d
fusion-blueprint                    1/1     1            1           33d
fusion-deployments                  1/1     1            1           33d
fusion-drift                        1/1     1            1           33d
fusion-execution-scheduler          1/1     1            1           33d
fusion-mgmtworker                   1/1     1            1           33d
fusion-nginx                        1/1     1            1           33d
fusion-plugins                      1/1     1            1           33d
fusion-rest-service                 1/1     1            1           33d
fusion-secrets                      1/1     1            1           33d
fusion-stage-backend                1/1     1            1           33d
hzp-api-gateway-svc                 1/1     1            1           33d
hzp-attestation-svc                 1/1     1            1           33d
hzp-catalog-svc                     1/1     1            1           33d
hzp-certificate-svc                 1/1     1            1           33d
hzp-compute-svc                     1/1     1            1           33d
hzp-datacollection-svc              1/1     1            1           33d
hzp-ece-firmwareupdate-svc          1/1     1            1           33d
hzp-eo-initialization-svc           1/1     1            1           33d
hzp-events-svc                      1/1     1            1           33d
hzp-flag-manager-svc                1/1     1            1           33d
hzp-fru-cru-svc                     1/1     1            1           33d
hzp-health-check-svc                1/1     1            1           33d
hzp-iam-policy-svc                  1/1     1            1           33d
```

```
hzp-iam-session-svc                1/1     1            1           33d
hzp-iam-webhook                    1/1     1            1           33d
```

3. Check the status of all deployments in the relevant namespace. The values in the `READY` column indicate the number of currently ready pods compared to the number of wanted ready pods. These numbers should match in a healthy environment.

```
# kubectl -n hzp get deployments
NAME                                        READY   UP-TO-DATE   AVAILABLE   AGE
helm-controller                             1/1     1            1           18h
source-controller                           1/1     1            1           18h
delco-controller                            1/1     1            1           18h
eo-chart                                    1/1     1            1           18h
hzp-delco-svc                               1/1     1            1           18h
hzp-iam-webhook                             1/1     1            1           18h
hzp-ingress-authz-svc-00001-deployment      1/1     1            1           18h
hzp-security-svc-00001-deployment           1/1     1            1           18h
```

4. If a deployment is unhealthy or requires further troubleshooting, view the logs for the deployment for any relevant debugging information.

```
# kubectl -n hzp logs deployment/<deployment_name> --tail=3
2023-03-29T12:50:53.301Z        INFO    EOSERVICES
component is ready   {"controller": "edgeorchestrator", "controllerGroup":
"hzp.dell.com", "controllerKind": "EdgeOrchestrator", "EdgeOrchestrator":
{"name":"eo","namespace":"hzp"}, "namespace": "hzp", "name": "eo", "reconcileID":
"9f4e9b7a-8890-4bce-8ae3-05c00ed6d5fa"}
2023-03-29T12:50:53.301Z        INFO    FRONTIERSERVICES component has
started reconciling      {"controller": "edgeorchestrator", "controllerGroup":
"hzp.dell.com", "controllerKind": "EdgeOrchestrator", "EdgeOrchestrator":
{"name":"eo","namespace":"hzp"}, "namespace": "hzp", "name": "eo", "reconcileID":
"9f4e9b7a-8890-4bce-8ae3-05c00ed6d5fa"}
2023-03-29T12:50:53.301Z        INFO    FRONTIERSERVICES component
is ready     {"controller": "edgeorchestrator", "controllerGroup":
"hzp.dell.com", "controllerKind": "EdgeOrchestrator", "EdgeOrchestrator":
{"name":"eo","namespace":"hzp"}, "namespace": "hzp", "name": "eo", "reconcileID":
"9f4e9b7a-8890-4bce-8ae3-05c00ed6d5fa"}
```

Troubleshoot pods

About this task

Users should generally focus troubleshooting efforts at the deployment level, as pods are members of a deployment. However, it is sometimes useful to troubleshoot an individual pod for targeted debugging purposes. Perform the following steps to troubleshoot pods:

Steps

1. Log in as `root` to the host on which Dell Automation Platform is deployed.
2. Check the status of all pods in the relevant namespace and look for pods that are not in a `Completed` or `Running` status.

   NOTE: The failed pod below is a normal failure. An initialization job is expected to fail while the application state converges, and is not cause for alarm.

```
# kubectl -n hzp get pods | grep -v 'Running\|Completed'
NAME                                READY   STATUS      RESTARTS      AGE
mist-ee-portal-admin-init-8h4pr     0/1     Error       0             10m
```

3. If a pod is unhealthy or requires further troubleshooting, view the logs for the pod for any relevant debugging information.

```
# kubectl logs -n hzp mist-ee-portal-admin-init-8h4pr --tail 3
  File "/mist.api/src/mist/api/clouds/controllers/main/base.py", line 374, in update
    raise BadRequestError({"msg": str(exc), "errors": exc.to_dict()})
mist.api.exceptions.BadRequestError: Bad Request: {'msg': "ValidationError
```

```
(Cloud.EdgeServiceCloud:e9946567637b4057a49d43780f289e0a) (Field is required:
['host'])", 'errors': {'host': 'Field is required'}}
```

4. Users can also describe the pod. The pod description is helpful when troubleshooting pods that are failing to schedule or are having problems pulling their image from the remote repository.

```
# kubectl -n hzp describe pod mist-ee-portal-admin-init-8h4pr
...
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  12m   default-scheduler  Successfully assigned hzp/mist-ee-
portal-admin-init-8h4pr to <host_name>
  Normal  Pulling    12m   kubelet            Pulling image "registry.example.com/
dell/mist/ee/api:9310d1e6200ec685c03299816a1add93eb46ea6b"
  Normal  Pulled     12m   kubelet            Successfully pulled image
"registry.example.com/dell/mist/ee/api:9310d1e6200ec685c03299816a1add93eb46ea6b" in
212.279429ms (212.290178ms including waiting)
  Normal  Created    12m   kubelet            Created container portal-admin-init
  Normal  Started    12m   kubelet            Started container portal-admin-init
```

Troubleshoot jobs

About this task

The Dell Automation Platform orchestrator deployment process creates several Kubernetes jobs to perform one-time configuration activities. A Kubernetes job fundamentally creates a short-lived pod to perform a specific task. Perform the following steps to troubleshoot jobs.

Steps

1. Log in as `root` to the host on which Dell Automation Platform is deployed.
2. Check the status of all jobs in the relevant namespace and look for jobs that have not run to completion. The `COMPLETIONS` column shows the number of actual completions compared to the number of expected completions. These numbers should match in a healthy environment.

```
# kubectl -n hzp get Jobs
NAME                          COMPLETIONS   DURATION   AGE
vault-bootstrap               1/1           2m1s       5d22h
factoryreset-init-job         1/1           118s       5d22h
factoryos-init-job            1/1           2m         5d22h
hostos-init-job               1/1           107s       5d22h
owner-to2-setup               1/1           7s         5d22h
hzp-certificate-svc-job       1/1           27s        5d22h
mist-ee-migrations            1/1           15s        5d21h
mist-ee-opensearch-manage     1/1           42s        5d21h
mist-ee-portal-admin-init     1/1           18s        5d21h
```

3. If a job is unhealthy or requires further troubleshooting, view the logs for the job for any relevant debugging information.

```
# kubectl -n hzp logs jobs/vault-bootstrap
Defaulted container "vault-bootstrap-container" out of: vault-bootstrap-container,
init-vault (init)
{"level":"debug","component":"hzp-vault-
bootstrap","time":"2023-06-15T14:26:29Z","message":"Got secret \"vault-secret\"."}
```

4. If investigation indicates an issue with the pod itself, such as a problem scheduling the pod, then perform the pod troubleshooting steps from Troubleshoot pods.

Troubleshoot services

About this task

The Dell Automation Platform orchestrator deployment creates several Kubernetes services. Some of these services are for internal consumption by resources within the Kubernetes cluster, and other services are designed for external access. Perform the following steps to troubleshoot services.

Steps

1. Log in as `root` to the host on which Dell Automation Platform is deployed.
2. List services in the relevant namespace:

```
# kubectl -n istio-system get services
NAME                        TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)
               AGE
istiod                      ClusterIP      10.43.200.5     <none>
15010/TCP,15012/TCP,443/TCP,15014/TCP        18m
istio-ingressgateway        LoadBalancer   10.43.181.0     192.168.28.158
15021:32465/TCP,80:32097/TCP,443:32362/TCP   18m
internal-ingress-gateway    ClusterIP      10.43.106.242   <none>           80/TCP
               18m
hzp-internal-gateway        ClusterIP      10.43.3.253     <none>           80/TCP
               18m
```

3. List the infrastructure assets for the services. Ensure that all relevant services have assets that are associated with them:

```
# kubectl -n istio-system get endpoints
NAME                        ENDPOINTS
      AGE
istiod                      10.42.0.13:15012,10.42.0.13:15010,10.42.0.13:15017 + 1
more...   20m
istio-ingressgateway        10.42.0.14:15021,10.42.0.14:8080,10.42.0.14:8443
      20m
hzp-internal-gateway        10.42.0.34:8081
      19m
internal-ingress-gateway    10.42.0.34:8081
      19m
```

4. Describe the service and review its events for anomalies:

```
# kubectl -n istio-system describe service istio-ingressgateway
Name:                     istio-ingressgateway
...
Events:
  Type    Reason                Age   From                Message
  ----    ------                ----  ----                -------
  Normal  EnsuringLoadBalancer  21m   service-controller  Ensuring load balancer
  Normal  AppliedDaemonSet      21m   service-controller  Applied LoadBalancer
DaemonSet kube-system/svclb-istio-ingressgateway-75577d7c
  Normal  UpdatedLoadBalancer   21m   service-controller  Updated LoadBalancer with
new IPs: [] -> [192.168.28.158]
```

5. If issues are identified with the deployments or pods that support a service, then troubleshoot those resources according to the previous tasks.

Common problems and solutions

This section describes common issues that may be encountered when deploying Dell Automation Platform orchestrator and includes recommended resolutions.

Docker fails to resolve the container registry

Problem: A failure is encountered immediately when running the `install-upgrade.sh` script due to a name resolution failure.

```
CR preparation is complete!
  FQDN: 192.168.28.158
WARNING! Using --password via the CLI is insecure. Use --password-stdin.
Error response from daemon: Get "https://registry.example.com/v2/": dial tcp: lookup
registry.example.com: Temporary failure in name resolution
Loading EO loader image locally
Loaded image: hzp/eo-loader:0.1.475
Pushing EO Loader image to Repository : registry.example.com/dell/
The push refers to repository [registry.example.com/dell/hzp/eo-loader]
Get "https://registry.example.com/v2/": dial tcp: lookup registry.example.com: Temporary
failure in name resolution
Failed to push EO Loader image to Repository: registry.example.com/dell/
```

Solution: The Docker daemon is unable to resolve the name of the container registry. Ensure that the container registry is a valid FQDN. Or, create the necessary entry in the hosts file on the system, and restart the Docker daemon.

```
# echo "192.168.28.192 registry.example.com" >> /etc/hosts
# systemctl restart docker
```

Docker fails with a certificate warning when uploading the Dell Automation Platform orchestrator image

Problem: A failure is encountered immediately when running the `install-upgrade.sh` script due to a certificate error.

```
Config map created
CR preparation is complete!
  FQDN: 192.168.28.158
WARNING! Using --password via the CLI is insecure. Use --password-stdin.
Error response from daemon: Get "https://registry.example.com/v2/": x509: certificate
signed by unknown authority
```

Solution: The Docker daemon does not recognize a self-signed registry certificate. Ensure that the certificate exists and the system trust store is updated. Restart the Docker daemon to recognize the certificate change. The following solution assumes that the registry CA certificate exists on the local system as `registry.crt`.

```
# cp ./registry.crt /usr/local/share/ca-certificates/registry.crt
root@native-edge:~# update-ca-certificates -f
Clearing symlinks in /etc/ssl/certs...
done.
Updating certificates in /etc/ssl/certs...
rehash: warning: skipping ca-certificates.crt,it does not contain exactly one
certificate or CRL
rehash: warning: skipping dellca2018-bundle.pem,it does not contain exactly one
certificate or CRL
126 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
# systemctl restart docker
```

Installation fails due to FailedMount error

Problem: Installation of Dell Automation Platform orchestrator fails due to a FailedMount error from the vault pod. This error is due to a missing configuration in the environment setup. The error message is displayed with a message similar to the following:

```
MountVolume.MountDevice failed for volume "pvc-5a9efa80-61a5-4b57-8425-30f3332d0faa"
: rpc error: code = Internal desc = format
of disk "/dev/longhorn/pvc-5a9efa80-61a5-4b57-8425-30f3332d0faa" failed:
type:("ext4") target:("/var/lib/kubelet/plugins/kubernetes.io/csi/driver.longhorn.io/
047e7c20e25053120907cd4bba76fa08cbcfd91320e12483c4bda42aeca7b15f/globalmount") options:
("defaults") errcode:(exit status 1) output:(mke2fs 1.46.4 (18-Aug-2021)
/dev/longhorn/pvc-5a9efa80-61a5-4b57-8425-30f3332d0faa is apparently in use by the
system; will not make a filesystem here!
```

Solution: If installation fails, add Longhorn block devices to the `multipathd` service, as described in the Troubleshooting: `MountVolume.SetUp failed for volume` due to multipathd on the node page of the Longhorn site.

Docker credentials are stored unencrypted

Problem: The password and other credentials are stored without encryption in the config JSON file. An error message is displayed with the following text (or something similar):

```
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store
Login Succeeded
```

This happens when no credential store has been configured. By default, docker stores any credentials in `config.json` and shows this warning message during docker login.

Solution: To configure the credentials store, log out from docker and set the following in `$HOME/.docker/config.json`.

```
{
  "credsStore": "osxkeychain" // “osxkeychain” on macOS, “wincred” on windows, and
“pass” on Linux
}
```

If the credential store is not initialized, perform the following steps to configure the password store as docker credstore.

1. Download, extract, make executable, and move to make it available to $PATH:

   ```
   $ wget https://github.com/docker/docker-credential-helpers/releases/download/v0.6.4/
   docker-credential-pass-v0.6.4-amd64.tar.gz
   $ tar -xf docker-credential-pass-v0.6.4-amd64.tar.gz
   $ chmod +x docker-credential-pass
   $ mv docker-credential-pass /usr/local/bin
   ```

2. Change the directory to system root:

   ```
   $ sudo su
   $ cd /root
   ```

3. Edit the docker `config.json` file using nano or something similar:

   ```
   $ nano .docker/config.json
   ```

   The file should look like this:

   ```
   {
           "credsStore": "pass",
           "auths": {}
   }
   ```

   Generate a gpg key and copy the ID. It is a long character string in all caps (for example, KJSADUSN831RHAM):

   ```
   $ gpg --gen-keys
   Initialize pass
   $ pass init
   ```

4. Finally, log in to docker:

   ```
   $ sudo docker login
   ```

   Use the pass command to see the password store tree. Make sure you are in the system `/root` directory and have `sudo su` activated.

   ```
   $ pass
   ```

   It should look similar to this:

   ```
   └── docker-credential-helpers
      └── aHR0cHM7Ly9pbmrleC5kb2NrZXIuaW8vdjEv
         └── blitzdex27
   ```

6

Uninstall Dell Automation Platform

The following topics describe how to uninstall Dell Automation Platform.

Topics:

- Uninstall a Dell Automation Platform Kubernetes deployment
- Uninstall a Dell Automation Platform OVA deployment

Uninstall a Dell Automation Platform Kubernetes deployment

Before uninstallation of Dell Automation Platform, Dell Technologies highly recommends that users consider the following:

- Ensure that all important data is backed up. Uninstalling the platform and orchestrator may result in the loss of configuration settings, logs, and other critical data.
- Uninstalling the orchestrator can disrupt the services that Dell Automation Platform manages. Ensure to plan the uninstallation during a maintenance window to minimize any impact on operations.
- Verify that there are no active dependencies on the orchestrator. Other systems or applications relying on Dell Automation Platform services may be affected.

Perform the following simple commands to delete the namespace and thus uninstall Dell Automation Platform:

```
helm uninstall native-edge-orchestrator -n <ORCHESTRATOR_NAMESPACE>
helm uninstall portal -n <PORTAL_NAMESPACE>
kubectl delete ns <ORCHESTRATOR_NAMESPACE> <PORTAL_NAMESPACE>
```

Uninstall a Dell Automation Platform OVA deployment

Before uninstallation of Dell Automation Platform, it is highly recommended that users consider the following:

- Ensure that all important data is backed up. Uninstalling the orchestrator may result in the loss of configuration settings, logs, and other critical data.
- Fully offboard and clean up the orchestrator before beginning the uninstallation process.
- Uninstalling the orchestrator can disrupt the services that Dell Automation Platform manages. Ensure to plan the uninstallation during a maintenance window to minimize any impact on operations.
- Verify that there are no active dependencies on the orchestrator. Other systems or applications relying on Dell Automation Platform services may be affected.

Perform the following simple steps to uninstall Dell Automation Platform;

1. Shut down the Dell Automation Platform virtual machine.
2. To permanently uninstall the orchestrator, delete the VM from the disk.

A

Backup and restore

Before deploying Dell Automation Platform orchestrator, be sure to configure backups of data in case an outage or other corrupting event affects the platform.

Topics:

- Backing up and restoring a Kubernetes deployment
- Backing up and restoring an OVA deployment

Backing up and restoring a Kubernetes deployment

The following tasks describe how to back up Dell Automation Platform orchestrator to an S3-compatible storage bucket. The tools to perform the backup reside in the Kubernetes cluster. There are other options, but this example uses Longhorn, Dell Elastic Cloud Storage (ECS), and Velero to perform the backup.

- Backup requirements
- Sample backup and restore workflow
- Backup and restore setup
- Test the backup and restore
- Create a test restore job
- Create a scheduled backup

Backup requirements

Table 8. Backup requirements

Name	Description
Kubernetes	Dell Automation Platform orchestrator runs in a Kubernetes cluster.
Longhorn	Longhorn is a lightweight cloud-native distributed block-storage solution for Kubernetes. In addition, it enables backups to S3-compatible object storage.
Velero	Velero is a backup and restoration framework that is built for Kubernetes clusters. A server component runs in the cluster and can be interacted with through a command-line tool to initiate backups and restoration. Velero provides a set of Custom Resource Definitions (CRDs) in Kubernetes to handle and manage the backup and restore operations. It comes with a plug-in architecture.
Elastic Cloud Storage (ECS) or other S3-compatible object storage	Dell ECS is a modern, enterprise-grade, software-defined object storage platform that is designed for both traditional and next-generation workloads. It provides organizations with an on-premises alternative to public cloud solutions. It enables organizations to flexibly capture, store, protect, and manage unstructured data behind the enterprise firewall, with scalability that rivals the public cloud. **NOTE:** Any object storage solution that is compatible with the Amazon S3 protocol is appropriate for this role.

Sample backup and restore workflow

The following figure shows a sample workflow using Velero. It illustrates the following:

- Green arrows—backup workflow
- Orange dotted arrows—restore workflow

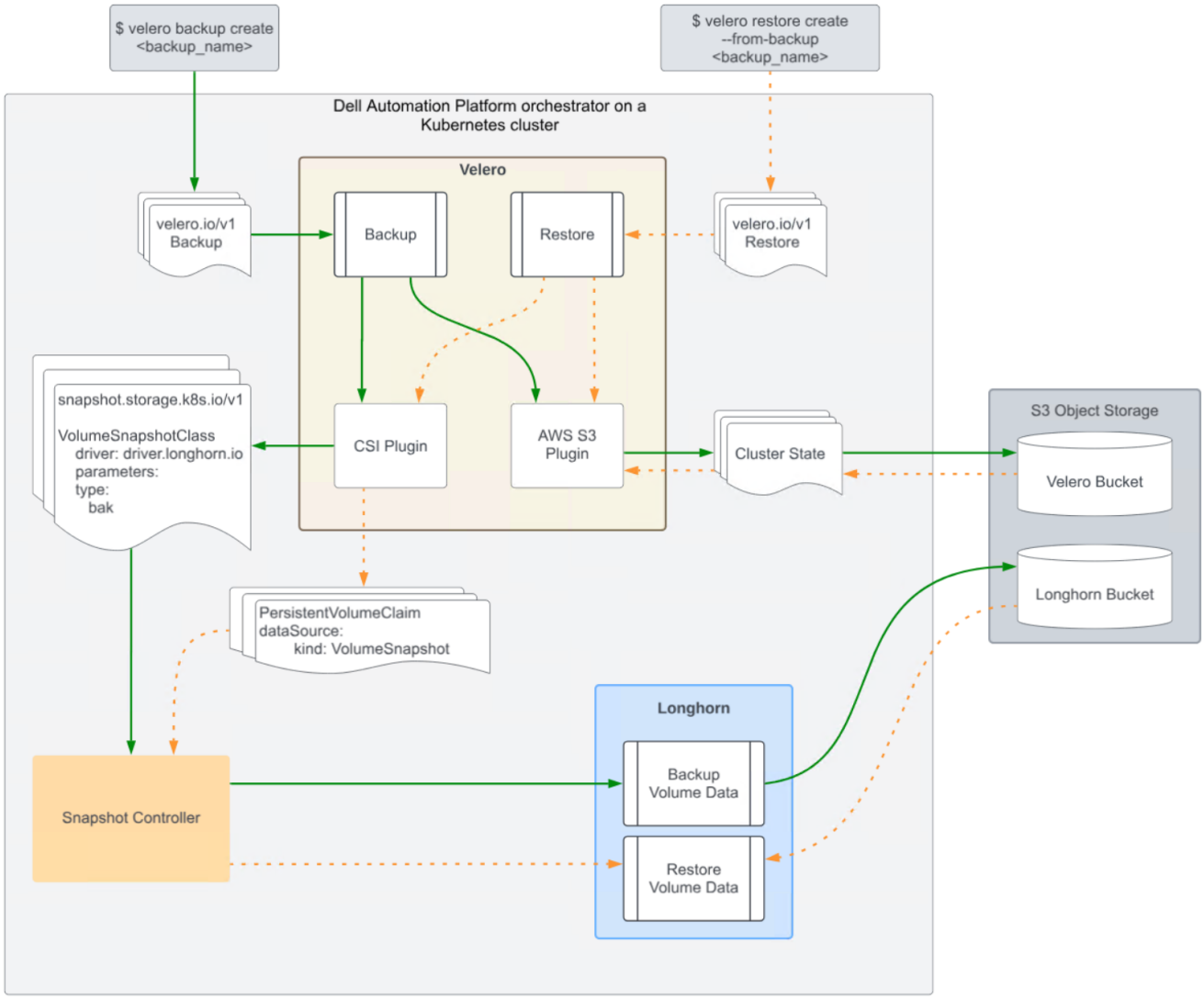

Figure 2. Sample backup and restore workflow

Velero uses the following plugins:

- `velero-plugin-for-aws`—For connection to Dell ECS https://github.com/vmware-tanzu/velero-plugin-for-aws
- `velero-plugin-for-csi`—For connection to Longhorn https://github.com/vmware-tanzu/velero-plugin-for-csi

In this example, install Longhorn and Velero in the Kubernetes cluster. Then configure both so that they can write backups to Dell ECS.

Use Dell ECS buckets for the following:

- Backup of the Dell Automation Platform orchestrator deployment YAML files
- Longhorn persistent volumes

Backup and restore setup

A Dell Automation Platform orchestrator deployment can use any S3-compatible object storage for backup and restore purposes. For the sake of demonstration, the following example uses Dell Elastic Cloud Storage (ECS). For more information about Dell ECS, see the Dell ECS Enterprise Object Storage product page.

Configure backup target in Longhorn

About this task

Perform the following steps to define the backup target as your object storage in Longhorn.

Steps

1. Access the Longhorn CLI.
2. Configure Dell ECS (or other object storage) as the Longhorn backup target.
   a. Generate a secret with values that Longhorn will use to access the Dell ECS environment.

   ```
   echo -n http://dellecs.company.edge:9020 | base64
   # aHR0cDovL21pbmlvLmVyNi1lZGdlLmxvY2FsOjkwMDA=
   echo -n admin | base64
   # YWRtaW4=
   echo -n supersecret | base64
   # RGVsbGVtYzEyMyE=
   ```

   The secret will be created in the `longhorn-system namespace`.

   b. Use the values in the following code to create the file, then apply by running the `kubectl apply -f <filename>` command.

   ```
   apiVersion: v1
   kind: Secret
     metadata:
       name: ecs-secret
       namespace: longhorn-system
     type: Opaque
     data:
       AWS_ACCESS_KEY_ID: YWRtaW4=
       AWS_SECRET_ACCESS_KEY: RGVsbGVtYzEyMyE=
       AWS_ENDPOINTS:
       aHR0cDovL21pbmlvLmVyNi1lZGdlLmxvY2FsOjkwMDA=
   ```

   c. Go to **Setting** > **General** in the Longhorn user interface and use the following format to configure the **Backup Target** endpoint URL:

   `s3://bucket_name@region/folder`

   For **Backup Target Credential Secret**, use the secret that was created in the previous substep.

3. Use the following commands to create the Container Storage Interface (CSI) backups. Kubernetes needs a snapshot controller and the CSI snapshot Custom Resource Definitions (CRD).

   - ```
     kubectl -n kube-system create -k "github.com/kubernetes-csi/external-snapshotter/client/config/crd?ref=release-6.3"
     ```
   - ```
     kubectl -n kube-system create -k "github.com/kubernetes-csi/external-snapshotter/deploy/kubernetes/snapshot-controller?ref=release-6.3"
     ```

   The backup option is enabled for the volumes.

4. Create a VolumeSnapshotClass for snapshots using the Longhorn driver.
   a. Under `labels:`, confirm inclusion of `velero.io/csi-volumesnapshot-class: "true"`.
   b. Under `parameters:`, specify `type` as **`bak`**, which tells Longhorn to make a backup.

   ```
   ```yaml
   kind: VolumeSnapshotClass
   apiVersion: snapshot.storage.k8s.io/v1
   ```
   ```

```
metadata:
  name: longhorn-snapshot-vsc
  labels:
    velero.io/csi-volumesnapshot-class: "true"          #confirm this line
driver: driver.longhorn.io
deletionPolicy: Delete
parameters:
  type: bak                                             #edit this line
```

Next steps

After you deploy and configure Dell Automation Platform orchestrator, proceed to Install Velero.

Install Velero

Prerequisites

Some type of S3-compatible object storage must be installed and configured. This example uses Dell ECS.

Users must install Velero from a Linux VM with access to the Kubernetes cluster.

Before you install Velero, complete the following tasks:

- Configure backup target in Longhorn.
- Deploy and configure the Dell Automation Platform as descibed in this guide.

About this task

Perform the following steps to install Velero.

Steps

1. Download and install the Velero CLI on a Linux VM with access to the Kubernetes cluster.

```
curl -LO https://github.com/vmware-tanzu/velero/releases/download/v1.12.0/velero-
v1.12.0-linux-amd64.tar.gz
tar zxvf velero-v1.12.0-linux-amd64.tar.gz
sudo mv velero-v1.12.0-linux-amd64/velero /usr/local/bin/velero
```

NOTE: See the official Velero documentation for a complete list of installation options.

2. Run the `velero` version to test that the Velero CLI installation worked correctly.
3. Create a credentials file specific to Velero for access to Dell ECS (or other object storage), for example `credentials-velero`.

```
aws_access_key_id = admin
aws_secret_access_key = myCompany123!
```

4. Set up Velero using the CLI with the `velero install` command.

```
velero install \
    --provider aws \
    --bucket <bucket_name> \
    --secret-file ./credentials-velero \
    --features=EnableCSI \
    --plugins=velero/velero-plugin-for-aws:v1.7.1,velero/velero-plugin-for-csi:v0.5.1
\
    --use-volume-snapshots=true \
    --backup-location-config
region=<region>,s3ForcePathStyle="true",s3Url=<backup_target_url>
```

NOTE: Replace *<bucket_name>*, *<region>*, and *<backup_target_url>* with your specific values.

The `Output` status begins.

5. View the status until the installation is complete.

```
Output
CustomResourceDefinition/backuprepositories.velero.io: attempting to create resource
CustomResourceDefinition/backuprepositories.velero.io: attempting to create resource
client
...

Deployment/velero: attempting to create resource

Deployment/velero: attempting to create resource client

Deployment/velero: created

Velero is installed! check 'kubectl logs deployment/velero -n velero' to view the
status.
```

Test the backup and restore

About this task

Perform the following steps to test the backup and to test a restore.

Steps

1. Create a backup, for example `dap-orchestrator-disaster-recovery-test`.
2. Exclude nonessential namespaces, such as `kube-system`.

```
velero create backup dap-orchestrator-disaster-recovery-test  \
--include-namespaces=hzp,portal \
--include-cluster-resources=true --csi-snapshot-timeout=30m
```

3. Use a command to verify that the backup is successful, for example `velero backup describe dap-orchestrator-disaster-recovery -test`.

```
Output...
velero backup describe dap-orchestrator-disaster-recovery-test
Name:         dap-orchestrator-disaster-recovery -test
Namespace:    velero
Labels:       velero.io/storage-location=default
Phase:  Completed
Namespaces:
  Included:  *
  Excluded:  kube-system
Resources:
  Included:        *
  Excluded:        <none>
  Cluster-scoped:  included
Label selector:  <none>
Storage Location:  default
Velero-Native Snapshot PVs:  auto
TTL:  720h0m0s
CSISnapshotTimeout:    30m0s
ItemOperationTimeout:  1h0m0s
Hooks:  <none>
Backup Format Version:  1.1.0
Started:    2023-09-06 14:10:14 +0000 UTC
Completed:  2023-09-06 14:25:18 +0000 UTC
Expiration:  2023-10-06 14:10:14 +0000 UTC
Total items to be backed up:  2007
Items backed up:              2007
Velero-Native Snapshots: <none included>
```

4. Check the `dap-orchestrator-disaster-recovery-bucket` to ensure that the backup files are stored.

5. Check the Longhorn bucket to ensure that the backup files are stored.

Create a test restore job

About this task

To test your backup setup, create an outage or other scenario so that you can restore the data on your Dell Automation Platform orchestrator.

Perform the following steps to simulate a disaster recovery.

Steps

1. Simulate a loss scenario by deleting the Dell Automation Platform orchestrator or some resources from it (for example, StatefulSets or secrets).
2. Use Velero to create a restore job, for example `dap-orchestrator-restore-test`.

```
velero restore create dap-orchestrator-restore-test --from-backup dap-orchestrator-
disaster-recovery-test
Restore request "dap-orchestrator-disaster-recovery-test" submitted successfully.
Run 'velero restore describe dap-orchestrator-disaster-restore-test' or 'velero
restore logs dap-orchestrator-disaster-restore-test' for more details.
```

3. Verify that the restore is successful with `velero describe restore nativeedge-orchestrator-restore-test`.
   The Output begins to run.

```
velero restore describe dap-orchestrator-disaster-restore-test

Name:         dap-orchestrator-restore-test
Namespace:    velero
Labels:       <none>
Annotations:  <none>
Phase:                       Completed
Total items to be restored:  1621
Items restored:              1621
Started:    2023-08-24 14:12:46 +0000 UTC
Completed:  2023-08-24 14:15:47 +0000 UTC
Warnings:   <none>
Errors:  <none>
Backup:  dap-orchestrator-disaster-recovery-test
Namespaces:
  Included:  all namespaces found in the backup
  Excluded:  <none>
Resources:
  Included:        *
  Excluded:        nodes, events, events.events.k8s.io, csinodes.storage.k8s.io,
volumeattachments.storage.k8s.io
  Cluster-scoped:  auto
Namespace mappings:  <none>
Label selector:  <none>
Restore PVs:  auto
Existing Resource Policy:   <none>
ItemOperationTimeout:       1h0m0s
Preserve Service NodePorts:  auto
```

4. Verify that the deleted Dell Automation Platform orchestrator resources are restored.

Create a scheduled backup

About this task

Perform the following steps to create the scheduled backup. This example uses seven days for the scheduled backup.

Steps

1. Enter the following to create a scheduled daily backup that lasts for seven days.

```
velero schedule create dap-orchestrator-disaster-recovery-daily --schedule "0 7 * *
*" \
--include-namespaces=hzp,portal --include-cluster-resources=true \
--csisnapshot-timeout=30m --ttl 168h0m0s
```

2. After a few days, check the backups in Velero to confirm that the system has created them according to the schedule.

```
velero get schedule dap-orchestrator-disaster-recovery-daily

NAME                                                STATUS    CREATED
      SCHEDULE    BACKUP TTL   LAST BACKUP   SELECTOR   PAUSED
dap-orchestrator-disaster-recovery-daily   Enabled   2023-08-22 15:01:41 +0000 UTC
0 7 * * *   0s           5h ago        <none>     false
```

3. Check Dell ECS (or other object storage) to confirm that the backup files are stored.

Backing up and restoring an OVA deployment

The following tasks guide users through taking and restoring a snapshot of the Dell Automation Platform virtual appliance on an vSphere client:

- Take a snapshot
- Restore a snapshot

Take a snapshot

Follow these steps to take a snapshot of the Dell Automation Platform virtual appliance.

About this task

To take a snapshot of a Dell Automation Platform virtual appliance:

Steps

1. Log in to the VMware vSphere client.
2. Follow the procedure in the VMware document Take Snapshots of a Virtual Machine.

   CAUTION: When taking a snapshot of the Dell Automation Platform virtual appliance, do not choose the *Include virtual machine's memory* option. Doing so causes Dell Automation Platform to fail to fully start.

Results

The snapshot is successful.

Restore a snapshot

Follow these steps to restore a snapshot of the Dell Automation Platform virtual appliance.

About this task

To restore a snapshot of a Dell Automation Platform virtual appliance:

Steps

1. Log in to the VMware vSphere client.
2. Follow the procedure in the VMware document Restoring snapshots.

Results

The snapshot is successfully restored.

B

References

Topics:

- Dell Technologies documentation
- Other documentation
- Contacting support

Dell Technologies documentation

The following Dell Technologies documentation provides additional and relevant information. Access to these documents depends on your login credentials. If you do not have access to a document, contact your Dell Technologies representative.

On the **Manuals & Documents** tab of the Dell Automation Platform Support Site:

- Dell Automation Platform—Administration Guide
- Dell Automation Platform—Pre-deployment Guide
- Dell Automation Platform—Deployment Guide
- Dell Automation Platform—Security Configuration Guide
- Dell Automation Platform—Release Notes
- Dell Automation Platform—Simple Support Matrix
- Dell Automation Platform Event Codes—Reference Guide
- Dell Automation Platform Blueprint—Quick Start Guide

Dell My Account

Dell Elastic Cloud Storage

Other documentation

The following documentation provides additional and relevant information:

- Kubernetes - Documentation
- Install Docker Engine on Ubuntu
- Installing Helm
- Longhorn - Documentation
- Velero - Documentation

Contacting support

For technical support, go to the Dell support site or call (USA) 1-800-945-3355.